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Final Project Report

Introduction

Single-cell RNA sequencing (scRNA-seq) has become a fundamental tool for studying cellular heterogeneity, allowing us to identify distinct cell types, characterize cellular states, and investigate disease mechanisms at high resolution (Satija et al., 2015). An important early step in scRNA-seq analysis is quality control (QC), where low-quality cells (such as empty droplets, damaged cells, or multiplets) are removed before downstream analyses like clustering and differential expression (Luecken and Theis, 2019). Poor quality control can distort results, introducing noise that obscures biological signals or removing biologically meaningful cells, which ultimately leads to misleading conclusions (Kiselev et al., 2019).

The paper highlighted that common QC approaches rely on global, rule-based filtering using metrics such as detected genes (nGenes), total counts (nCounts), and mitochondrial read percentage (Hao et al., 2021). These “standard cutoff” methods apply fixed thresholds across all cells, implicitly assuming QC metrics are comparable across cell types. However, this ignores biological variability: different cell types naturally exhibit distinct transcriptional and metabolic profiles, such as lower gene complexity or higher mitochondrial content (Kiselev et al., 2019). These limitations motivate more adaptive, data-driven QC methods.

Subramanian et al. (2022) address this by introducing ddqc, a cluster-based QC framework that performs filtering within transcriptionally similar cell groups rather than across the entire dataset. By leveraging clustering as a proxy for biological structure, ddqc enables QC thresholds to adapt to cell-type-specific distributions.

The main challenge motivating ddqc is distinguishing true biological variation from technical noise, especially in heterogeneous or disease contexts where cells may deviate from typical QC profiles. For example, in kidney fibrosis, cell types such as myofibroblasts may exhibit elevated transcriptional activity or altered gene expression that could be misclassified as low quality under standard filtering (Tang et al., 2021; Chen et al., 2024).

Our project reproduces the ddqc framework and evaluates how different QC strategies (ddqc, miQC, standard cutoff, and no QC) affect clustering outcomes and biological interpretability. We perform this comparison on a colon immune cell dataset used in Subramanian et al. (2022). The “no QC” condition serves as a baseline, while standard cutoff represents traditional filtering. miQC (Hippen et al., 2021) and ddqc provide complementary data-driven approaches (probabilistic and cluster-based respectively) allowing us to assess how QC choice influences clustering structure, noise, and identification of biologically meaningful populations.

We further extend the original work to address limitations of ddqc. First, we examine sensitivity to clustering resolution, which the original paper fixes at ~1.4 without evaluation. Prior literature shows clustering in scRNA-seq is highly sensitive to resolution (Kiselev et al., 2019; Fortunato and Barthélemy, 2007; Traag et al., 2019), which controls cluster granularity: lower values

merge populations, while higher values fragment them. Since ddqc operates within clusters, clustering inaccuracies may directly affect which cells are retained or removed. Therefore, we systematically vary resolution to assess ddqc's robustness.

Second, we evaluate the assumption that clusters correspond to biologically meaningful cell types. This assumption may not hold in complex or noisy datasets. We assess whether clusters generated under different QC methods align with known biological structure using clustering patterns and marker-based interpretation.

Finally, we analyze more complex biological settings, including healthy versus disease kidney data, where QC metrics may reflect underlying biology rather than technical artifacts. This allows us to evaluate whether data-driven approaches such as ddqc and miQC better preserve disease-relevant variation compared to standard filtering.

Overall, by systematically examining clustering parameters, biological context, and QC methods, we provide a more comprehensive assessment of how QC choices influence single-cell analysis and identify conditions under which current approaches may be improved.

Method

Original Paper Methods (Subramanian et al., 2022)

Subramanian et al. (2022) proposed ddqc as an adaptive, cluster-aware QC framework for scRNA-seq data. The pipeline begins with minimal filtering, removing cells with fewer than 100 detected genes or more than 80% mitochondrial reads to eliminate obvious low-quality cells. The remaining cells are log-normalized (log-TPX+1), 2,000 highly variable genes are selected, and PCA is performed using 50 components. A k-nearest neighbor graph (k = 20) is constructed, followed by Louvain clustering at resolution 1.4.

Within each cluster, adaptive cutoffs are computed as median ± 2 × MAD for four QC metrics (gene count, UMI count, mitochondrial fraction, and ribosomal fraction), with floor/ceiling constraints (minimum 200 genes, maximum 10% mitochondrial) to prevent overly permissive thresholds(Subramanian et al., 2022). Cells outside these cluster-specific bounds are removed before downstream analysis.

To compare the performance of ddqc to other methods, the paper compared ddqc to standard QC metrics (fewer than 200 genes or greater than 10% mitochondrial reads removed), no QC, and miQC (a probabilistic framework for QC metric calculation). For the standard QC, no QC, and ddqc, the joint comparison was run by labeling each barcode according to which methods retained it, clustering all non-"neither" barcodes jointly, and visualizing method-specific retention patterns on a shared UMAP.

For the miQC (Hippen et al., 2021) comparison to ddqc, the paper used two datasets: the Tabula Muris heart and aorta and human olfactory epithelium and examined the intersection of barcodes retained by each method.

Recreation

In our recreation, we compared four QC methods (no QC, standard QC, miQC, and ddqc) using immune colon cells from the Smillie et al. (2019) dataset, as in the original paper. Raw count matrices (20,529 genes × 210,614 cells) were loaded in R using ReadMtx, metadata was added, and immune cells were subsetted. A shared unfiltered Seurat object (colon_immune_raw.rds) was created as a common input across all QC branches.

For ddqc, we followed the pipeline described in the paper using the ddqcR package in Seurat. For miQC, we followed the published vignette: mitochondrial genes were identified by the “MT-” prefix, per-cell QC metrics were computed using addPerCellQC, and a two-component mixture model was fit and visualized with plotMetrics. Cells with posterior probability ≥ 0.75 of being low-quality were removed using filterCells.

For standard QC, fixed thresholds (fewer than 200 genes or greater than 10% mitochondrial reads) were applied following the Seurat PBMC vignette. For no QC, the full dataset (210,614 cells) was passed directly to downstream analysis without filtering.

All four conditions used identical downstream processing (log-normalization, 2,000 variable features, scaling, PCA, k-NN, Louvain clustering at resolution 1.4, and UMAP using the top 20 PCs), ensuring that observed differences were attributable solely to QC strategy.

<u>Extension 1: Disease State Generalization (Kidney Dataset)</u>
To evaluate ddqc and miQC on healthy versus disease data, we used IgA nephropathy (IgAN) kidney samples from Tang et al. (2021) (GEO: GSE171314). Healthy (GSM5222734, 6,438 cells) and disease samples (GSM5222730–GSM5222733, 14,132 cells) were loaded as separate Seurat objects with condition labels and barcode prefixes.

For ddqc, both conditions were merged into a single object (20,570 cells) and processed using the same pipeline as in the recreation. For miQC, each condition was processed separately following the vignette, with two additional parameters enabled (*keep_all_below_boundary* and *enforce_left_cutoff*) to prevent incorrect removal of low-mitochondrial cells and avoid U-shaped filtering boundaries, which are more common in disease data.

After filtering, each condition was processed through a consistent Seurat workflow (normalization, 2,000 variable features, scaling, PCA with 50 components, k-NN with k = 20, Louvain clustering at resolution 1.4, and UMAP using the top 20 PCs).

A comparison script was then used to compute cell retention per condition, quantify barcode-level overlap between ddqc and miQC (kept by both, ddqc only, miQC only, or removed by both), visualize QC metric distributions across retention groups using violin plots, and embed all pre-QC cells in a joint UMAP colored by retention category.

<u>Extension 2: Clustering Resolution Sensitivity</u>
To assess the sensitivity of ddqc to clustering granularity, we varied the Louvain resolution parameter across five values (0.4, 0.8, 1.3, 1.6, and 2.0) on the colon immune dataset, extending Subramanian et al. (2022) that fixes resolution at ~1.4 without sensitivity analysis. For each resolution, we applied the Seurat pipeline (normalization, 2,000 variable features, scaling,

PCA with 50 components, k-NN graph with k = 20), followed by Louvain clustering and UMAP visualization. ddqc filtering was then performed using fixed cluster-specific MAD-based thresholds. The pipeline was rerun on the filtered cells at the same resolution to ensure comparability. We evaluated ddqc performance by comparing pre- and post-filtering UMAPs, clustering structure, and the fraction of cells removed across resolutions.

Extension 3: Cluster Coherence via Independent Annotation

To assess whether ddqc clusters reflect biologically meaningful groupings rather than arbitrary partitions, we compared ddqc cluster assignments to cell type labels from two independent reference-based annotation tools: CellTypist (Immune_All_Low model) and SingleR (MonacoImmuneData).

Annotations were generated without prior knowledge of ddqc clusters and matched to cluster IDs via cell barcodes. For each resolution tested, a row-normalized cross-tabulation matrix was computed and visualized as a heatmap of annotation proportions per cluster. Cluster purity was defined as the proportion of cells within each cluster assigned the dominant annotation label.

To further evaluate consistency, MAD-based QC thresholds were recomputed using annotation labels instead of cluster IDs (2 × MAD on n_genes and percent_mito), and pass/fail decisions were compared cell-by-cell with the original ddqc results. Concordance between these approaches indicates that ddqc clusters align with biologically meaningful groupings and produce comparable adaptive QC thresholds.

GitHub repo: https://github.com/Tazein-S/Data_Driven_scQC_Benchmarking

Results

Original Paper Results (Subramanian et al., 2022):

For the paper's comparison between standard QC, no QC, and ddqc, the joint comparison showed that ddqc retained a median of 95.4% of input cells versus 69.4% under the standard cutoff. No cluster met the uniqueness threshold under the standard cutoff, while ddqc produced multiple biologically meaningful unique clusters. Of cells removed by ddqc, 83.19% were filtered due to mitochondrial thresholds and 6.2% due to gene complexity, confirming that ddqc's leniency is concentrated in mito-rich cell populations rather than uniformly low-quality cells (Subramanian et al., 2022).

For the miQC (Hippen et al., 2021) comparison to ddqc, the paper does not discuss their results for the comparison between miQC and ddqc in extensive detail. They state that ddqc and miQC retained all the clusters of the human Olfactory Epithelium dataset, with ddqc retaining more of the mito-rich olfactory epithelial cells. However, in the Tabula Muris mouse heart dataset, miQC completely removed the cardiomyocyte cluster, which is biologically significant because cardiomyocytes are essential parenchymal cells of the heart. In both of the datasets, ddqc was able to retain more cells, including those with more biological relevance.

Recreation Results

Within our recreation, we applied all four QC methods to the colon immune cells (Smillie et al., 2019), which had 210,614 cells before filtering. ddqc retained 210,066 cells (~600 removed), standard cutoff retained 198,586 cells (~12,000 removed), and miQC kept 196,050 cells (~14,000 removed), while no QC retained all cells. This is consistent with the pattern seen in the paper where ddqc retains the most cells. The biological significance of ddqc's additionally retained cells was explored further in the extension.

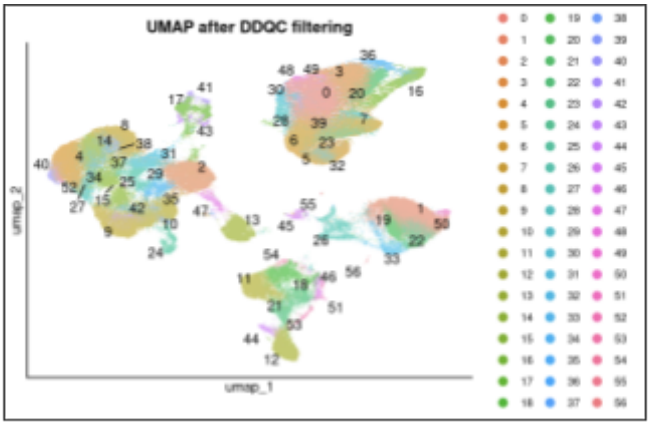

Fig 1. UMAP from ddqc

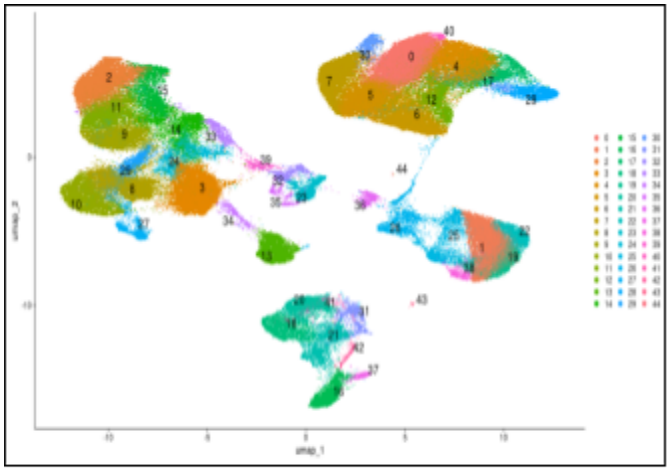

Fig 2. UMAP from Standard QC

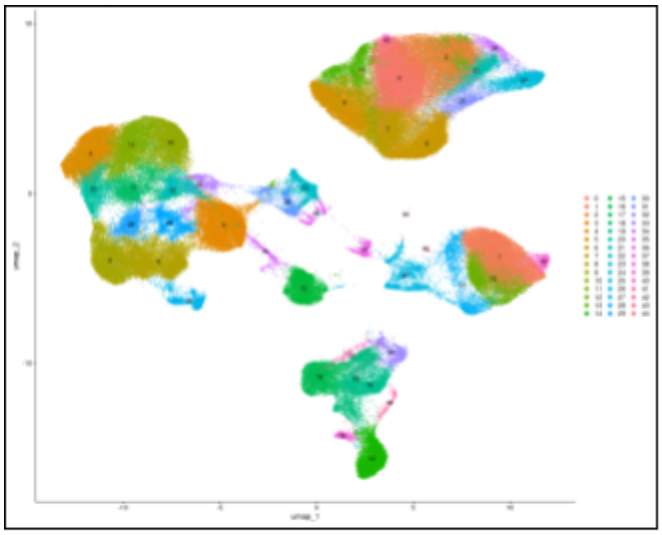

Fig 3. UMAP from No QC preprocessing

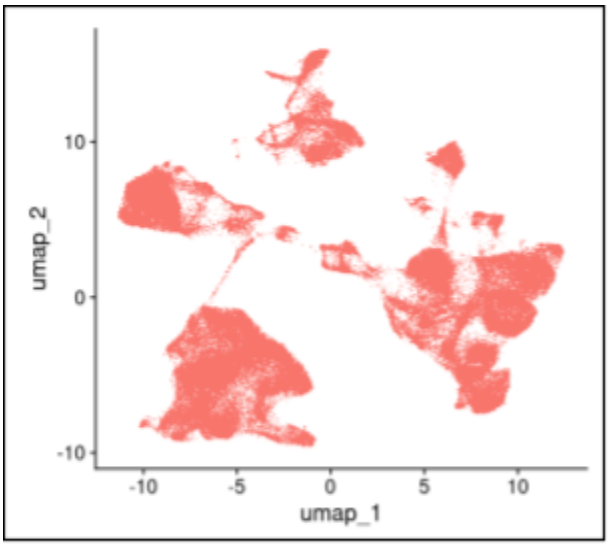

Fig 4. UMAP from miQC

For ddqc, the UMAP produced 34 compact, well-separated clusters, reflecting the method's ability to preserve biological heterogeneity while removing minimal cells. The standard QC UMAP produced 44 clusters, with structures divided into three broad regions: a dense top-right island of likely dominant cell types, a fragmented left island, and a separated bottom island.

Smaller isolated fragments suggest rare cell types, doublets, or poor-QC survivors, and a thin bridging cluster between islands may represent a transitional cell state. miQC produced a similar overall structure to standard QC but with fewer cells, reflecting its more aggressive filtering relative to ddqc. With no quality control, the UMAP shows diffuse cluster boundaries, elongated clusters, and bridging structures between populations, reflecting noise from retained low-quality cells that obscures biological signals and makes interpretation more difficult.

Extension 1: Disease State Generalization (Kidney Dataset)

When it came to running QC metrics on the healthy vs IgAN (disease) kidney samples, ddqc and miQC showed drastically different results. For the healthy kidney samples, both ddqc and miQC removed similar amounts of cells, with ddqc keeping 100% of the cells while miQC kept 97.86% of cells. However, when it came to the IgAN samples, ddqc kept 99.55% of cells while miQC only kept 36.16% of cells. This means that miQC removed over 8,800 cells from the disease condition.

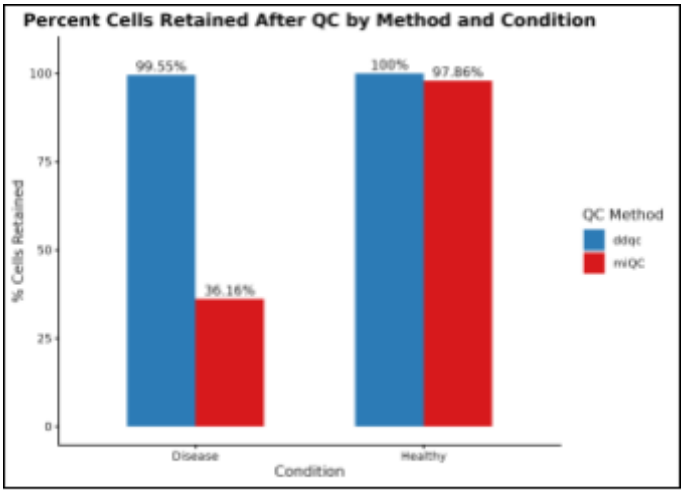

Fig 5. A bar chart showing the percentage retention rate of cells between the healthy and disease state using ddqc and miQC

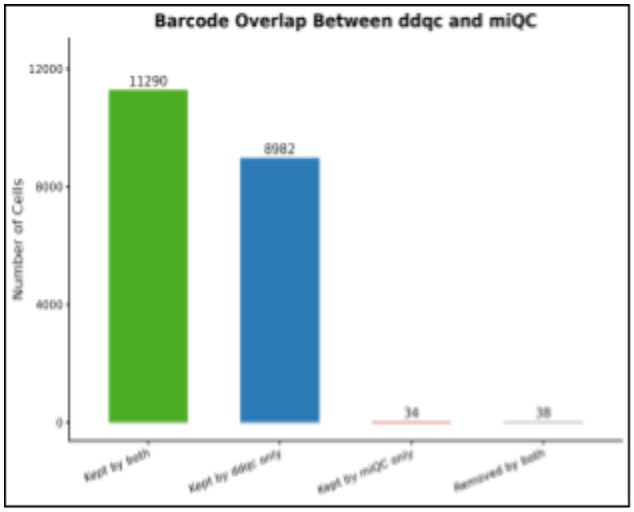

Fig 6. A bar chart showing the overlapping cell barcodes between ddqc and miQC

To examine which cells were removed, the comparison also calculated the barcode overlap between miQC and ddqc filtering of the IgAN samples. The barcode chart shows that 11,290 cells were kept by both methods, 8,982 were kept by ddqc only, just 34 were kept by miQC only, and 38 were removed by both. This is valuable, because it shows that miQC and ddqc did agree on a majority of cells that were retained in both methods, however, miQC was more aggressive in general by removing cells that ddqc retained.

This drastic divergence between miQC and ddqc filtering can be explained by looking at the gene count distributions and mitochondrial gene percentages across all the cells retained by ddqc, miQC, both and none of the methods. ddqc kept cells with high mitochondrial read percentages, while miQC seems to have a hard cut off of cells near mitochondrial percentage of 30%. However, the gene count distribution between both methods shows that both methods kept a similar amount of cells with low gene count distribution. This is significant because the high percentage of mitochondrial reads in the cells of disease samples aligns with what is

known about IgA nephropathy, where disease-driven inflammation and metabolic stress in kidney cells can elevate mitochondrial activity, which shows that ddqc is able to retain more biologically significant cells.

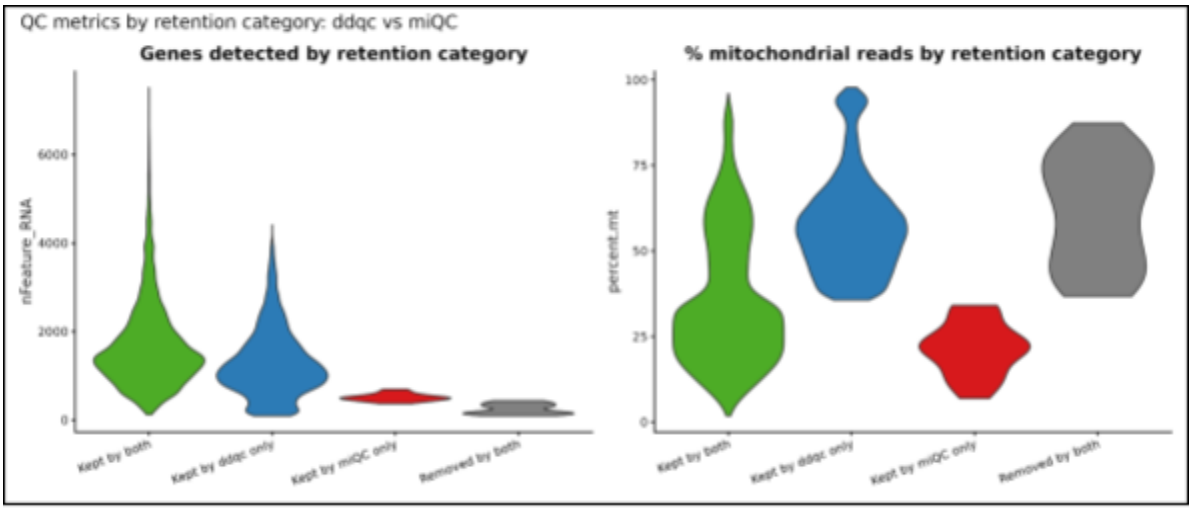

Fig 7. violin plots illustrating the number of genes detected by retention and % mitochondrial reads retention between ddqc and miQC

In addition, the UMAP shows that ddqc-retained cells are not noisy, but align with the same clusters of cells retained by miQC. In addition, the pre and post ddqc UMAP shows that the overall structure of clusters are preserved with both healthy and disease cells maintaining their relative organization across conditions.

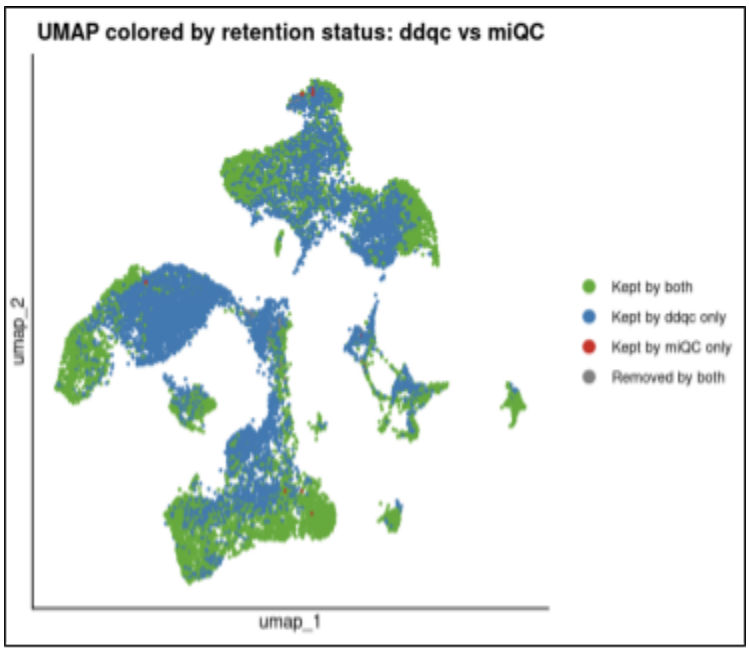

Fig 8. UMAP colored by retention status between ddqc and miQC

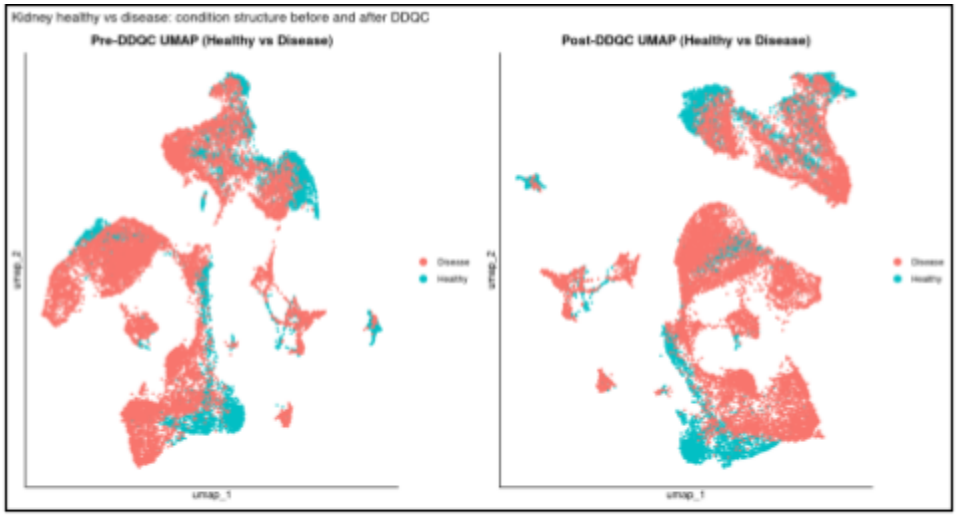

Fig 9. UMAP visualization of cell distribution and clustering structure in disease vs. healthy samples before and after ddqc

After comparing miQC and ddqc, it is clear that ddqc retains more cells of biological significance, which is valuable when analyzing disease samples that do not follow traditional scRNA behavior. This is essential, since ddqc still removes cells. However, it is less stringent than traditional QC or other methods such as miQC.

<u>Extension 2: Clustering Resolution Sensitivity</u>

We evaluated clustering structure across resolutions 0.4, 0.8, 1.3, 1.6, and 2.0, both before and after applying ddqc. As expected, increasing the resolution parameter led to a greater number of clusters, transitioning from broad groupings to finer partitions. At lower resolutions (0.4 to 0.8), clusters were large and often merged, while higher resolutions (1.6 to 2.0) produced smaller, more fragmented clusters.

Before ddqc, clustering across all resolutions appeared noisier, with diffuse boundaries and scattered cells, likely reflecting low-quality or stressed cells. This effect was more pronounced at higher resolutions, where over-fragmentation amplified noise and resulted in poorly defined clusters. After ddqc, clusters became more compact and clearly separated across all resolutions, indicating improved signal-to-noise.

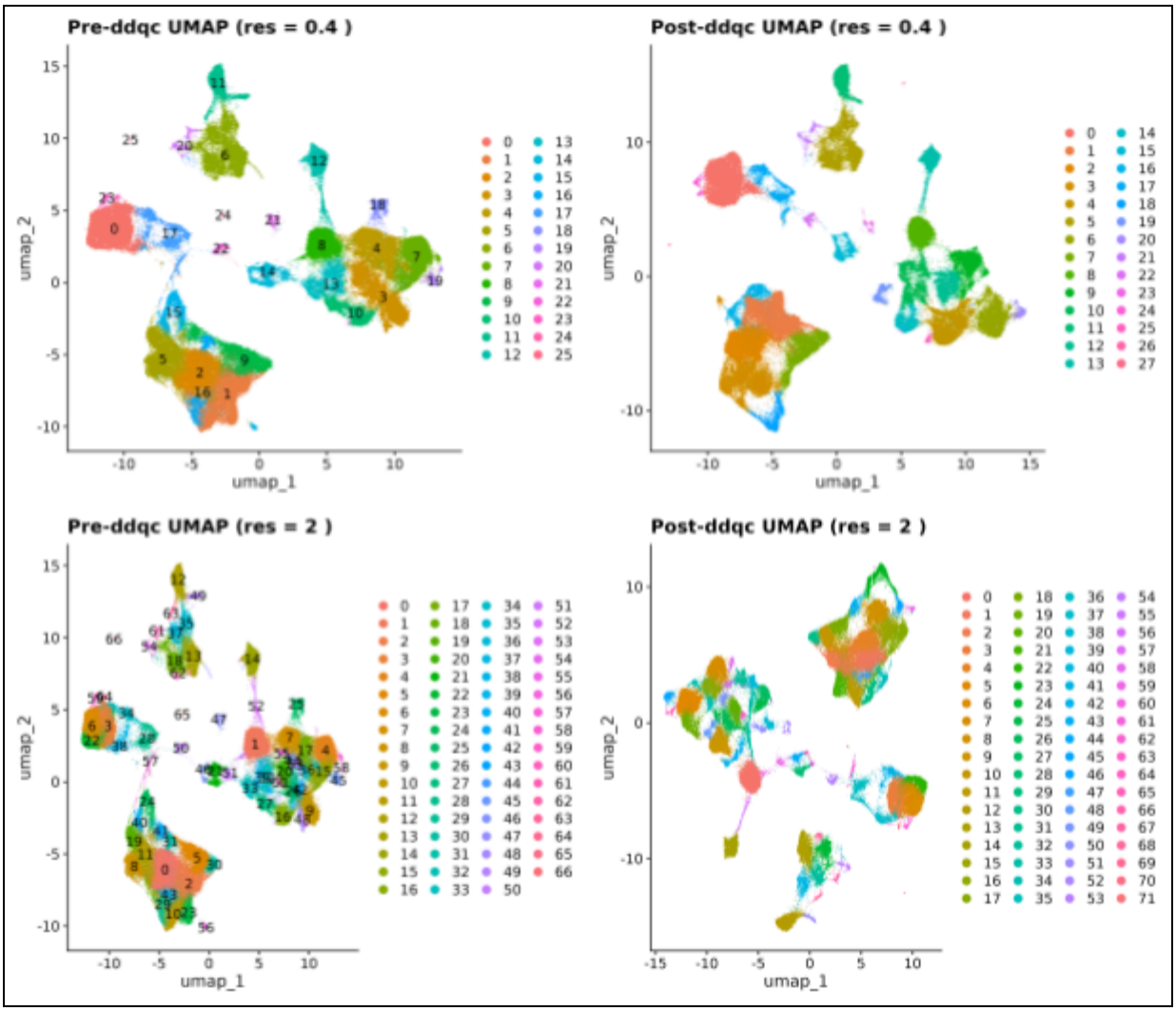

Fig 10. UMAP visualizations before and after ddqc across resolutions 0.4 and 2.0. ddqc improves cluster compactness and separation across levels of granularity, with the most interpretable structure observed at intermediate resolutions (1.3 to 1.5)

At intermediate resolutions (≈1.3 to 1.6), clustering was most interpretable, balancing separation of distinct populations with structural coherence. In contrast, at very high resolution (2.0), clusters were very fragmented even after QC. These findings are consistent with the original ddqc paper (resolution ≈1.4), while extending it by demonstrating that clustering resolution meaningfully influences outcomes.

Importantly, the proportion of cells removed by ddqc remained stable across resolutions (~15.9%–16.7%), indicating that filtering strength is robust to clustering granularity. However, while the number of cells removed is consistent, clustering structure and partitioning remain sensitive to resolution. Moderate resolutions provide the best balance between biological interpretability and granularity, highlighting that resolution tuning remains important even when using data-driven QC methods.

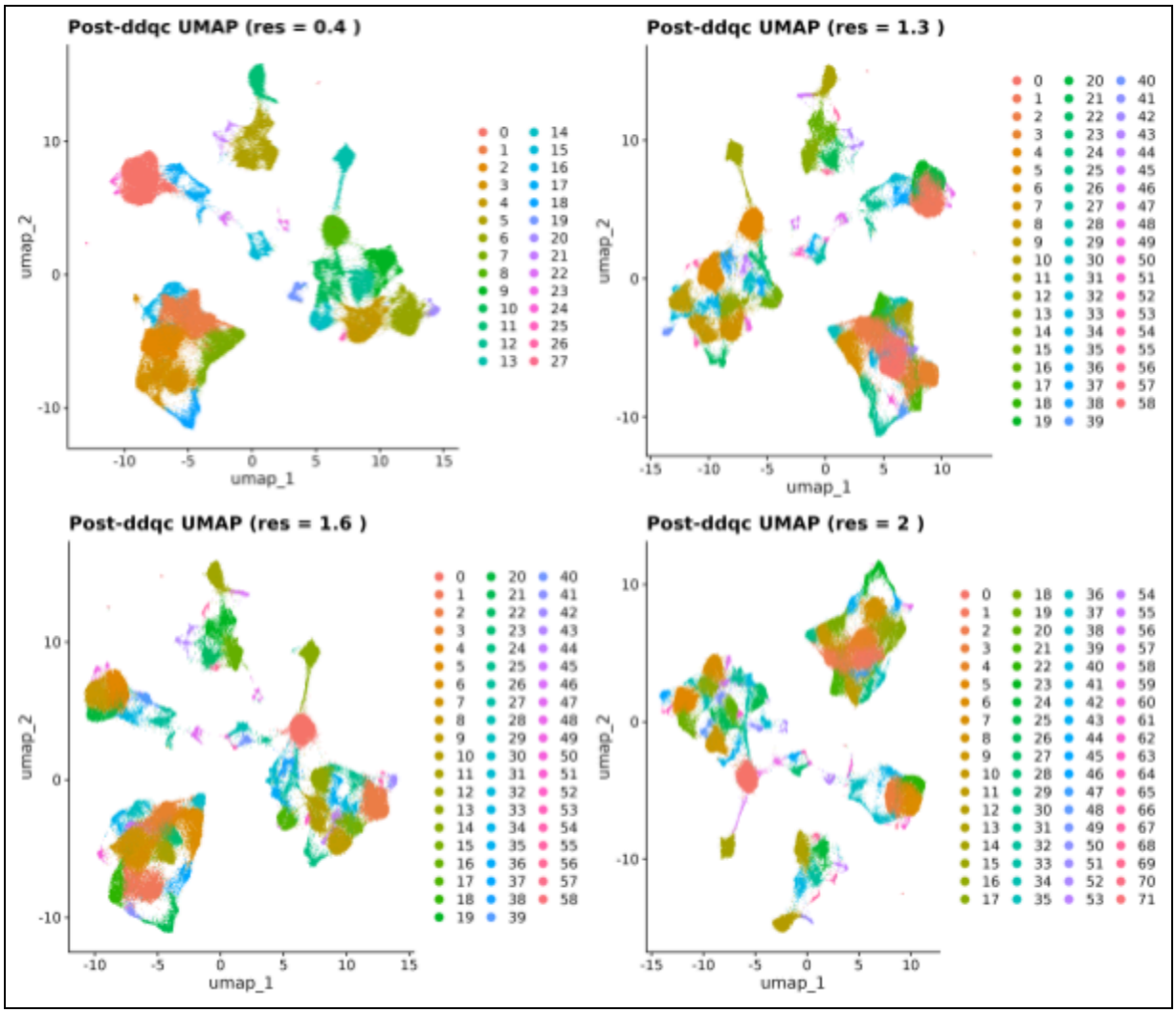

Fig 11. UMAP visualizations after ddqc across increasing resolution values (0.4 → 2.0), illustrating the transition from coarse to fine-grained clustering. Higher resolutions produce increased fragmentation despite improved noise reduction from ddqc.

Method	Concordance with CellTypist	Concordance with singleR
ddqc 0.4	0.8746	0.8764
ddqc 0.8	0.8764	0.8832
ddqc 1.3	0.8791	0.8792
ddqc 1.6	0.8727	0.8728
ddqc 2	0.8715	0.8715
Standard QC	1	1
miQC	0.9754	0.9777

Table 1. The concordance percentage between ddqc resolutions, miQC, and standard cutoff against CellTypist and singleR after MAD filtering

Extension 3: Cluster Coherence via Independent Annotation

Concordance measures agreement between ddqc clustering and reference-based annotations by comparing QC decisions after replacing cluster IDs with annotation labels and applying the same MAD-based filtering. dqc achieved ~87% concordance, with ~12–13% disagreement, while miQC exceeded 95% and standard QC reached 100%. This is more likely to be a potential bias of the QC approach. ddqc is an unsupervised method where mismatches are expected, and since it is a graph-based clustering, it may result in different QC boundaries.

Retention rates across cell types were broadly consistent across methods, with miQC and standard QC nearly overlapping. In contrast, ddqc showed slightly lower retention and more variability, indicating non-uniform, cluster-specific filtering that depends on local population structure.

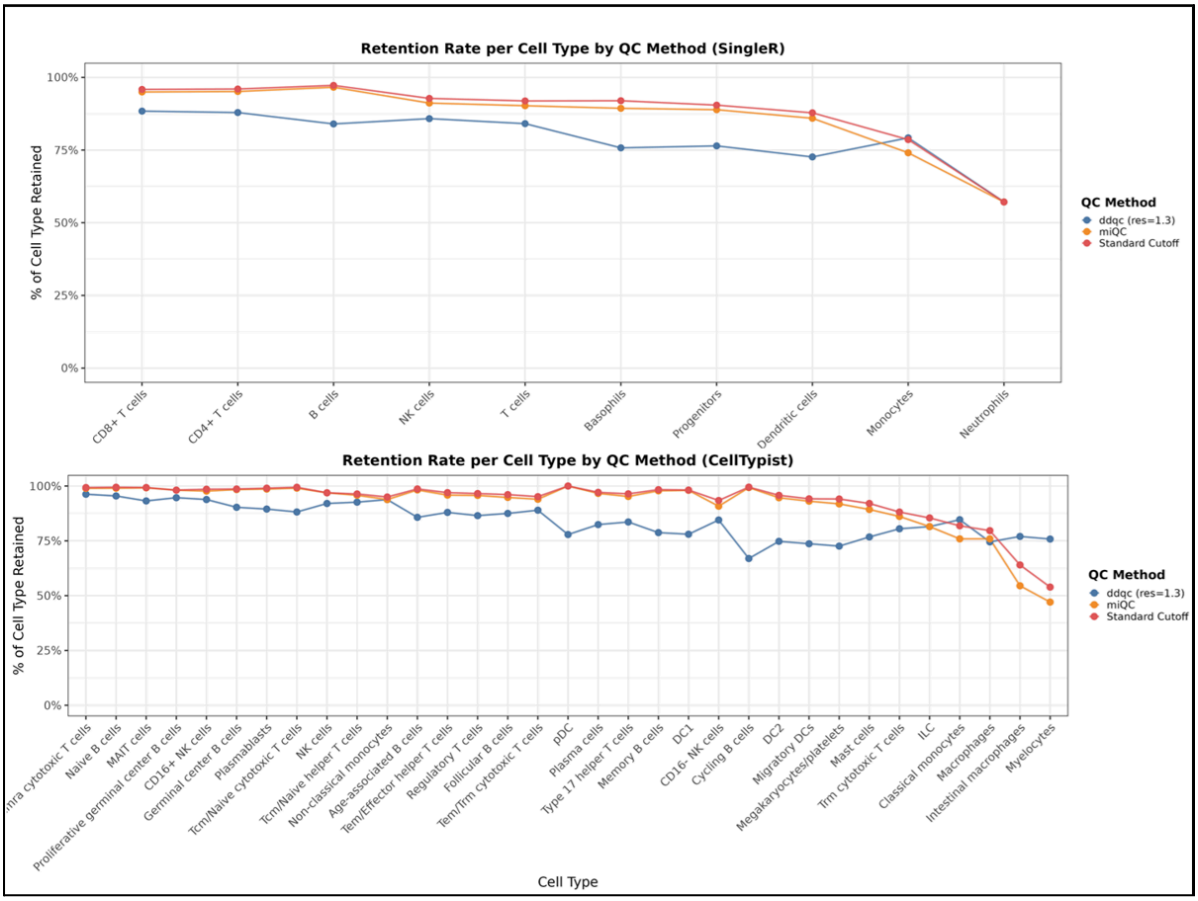

Fig 12. Line graph of retention rate per cell type between ddqc, miQC, and standard cutoff method using singleR (top) and CellTypist (bottom)

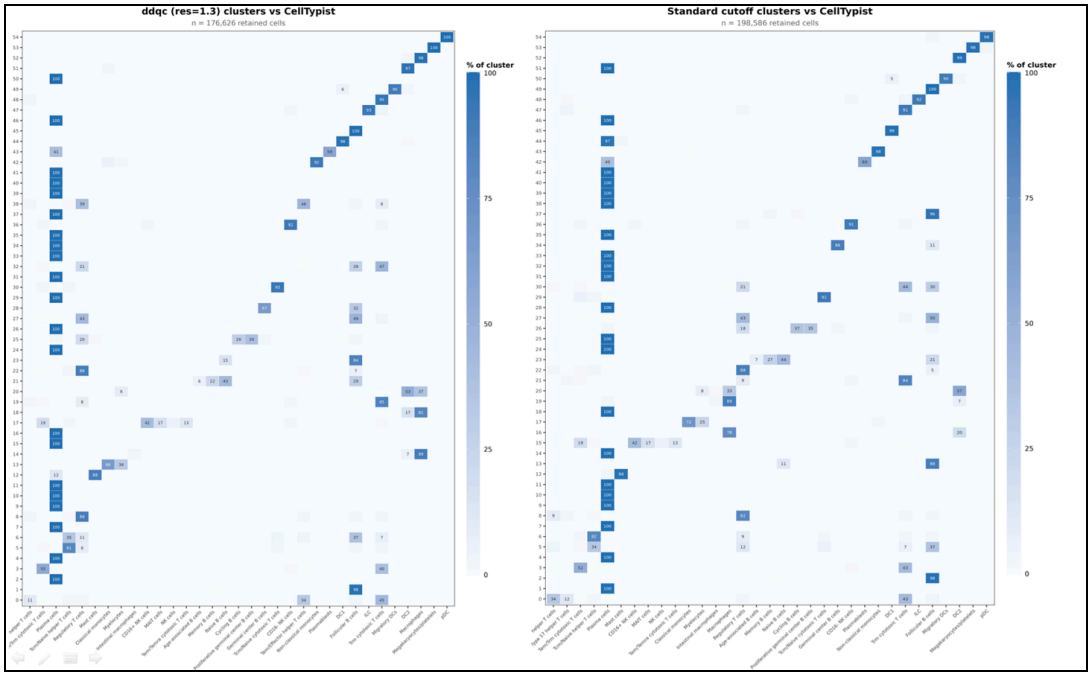

Fig 13. Heatmap generated from cross-tabulation of matching cell barcodes between ddqc clustering with resolution of 1.3 vs standard cutoff using cell type annotation (CellTypist)

Figure 13 visualizes the purity of each cluster generated from ddqc and standard cutoff QC methods, with the number of clusters on the vertical axis and cell types on the horizontal axis. Both of the QC processes resulted in 54 clusters, with the standard cutoff retaining more cells (198,586 cells) than ddqc (176,626) after MAD threshold filtering. An aligned diagonal across the heatmap represents high purity of each cluster, meaning that each cluster has biological uniqueness that matches the reference-based cell annotation. The top right corner of the plots demonstrates good alignment of cell type purity within a cluster, and the continuity along the diagonal line is better on the ddqc 1.3 resolution plot. Other than diagonal alignment, the repeating block fragment along the vertical axis is an indicator of multiple different clusters categorized as the same cell types by the annotation tools. Both heatmaps show that cells annotated as plasma cells cluster into distinct groups. This observed fragmentation could be arising from the biology of plasma cells, which are heterogeneous and may have subpopulations within the cell type.

Another possibility is that ddqc overclustered, resulting in fragmented groups of the same cell type. However, these interpretations remain tentative due to the lack of marker-based validation in the dataset. Nevertheless, the pattern is structured rather than random, and these results further confirm that clustering could introduce bias for some highly specialized types of cells. Notably, while this fragmentation is visible in both heatmaps, the repeating blocks mostly occur on the standard cutoff heatmap and are more scattered across the diagram. This possibly explains the concordance rate of standard QC being 100%, suggesting that this method may have higher within-cluster heterogeneity because it retained a broader range of cells due to a less stringent filtering process. As a result, clusters may appear consistent in terms of

annotation while still containing greater internal variability. Whereas for ddqc, each cluster had better within-cluster consistency despite the lower number of cells retained.

Additionally, among the resolutions used to evaluate clustering coherence, ddqc at 0.4 and 2 illustrate patterns of under- and overclustering, respectively. Resolution 2 produced more than double the number of clusters (66 vs. 25 at 0.4), while retaining slightly more cells (176,211 vs. 175,304). Although diagonal alignment at resolution 0.4 appears visually intact, agreement scores remain low, suggesting that broad clusters capture only small proportions of each cell type, consistent with underclustering that spreads cell populations too thinly across groups. These results indicate that resolution is not a neutral parameter but a hidden variable that interacts with cell type heterogeneity and should be considered in QC optimization rather than treated as a fixed default. More broadly, this sensitivity highlights a key limitation of ddqc: while it performs biology-aware QC within clusters, the clusters themselves are not biologically neutral. The granularity at which cell populations are resolved directly shapes QC thresholds, meaning ddqc's ability to preserve biological signals is inherently constrained by the same heterogeneity it aims to capture.

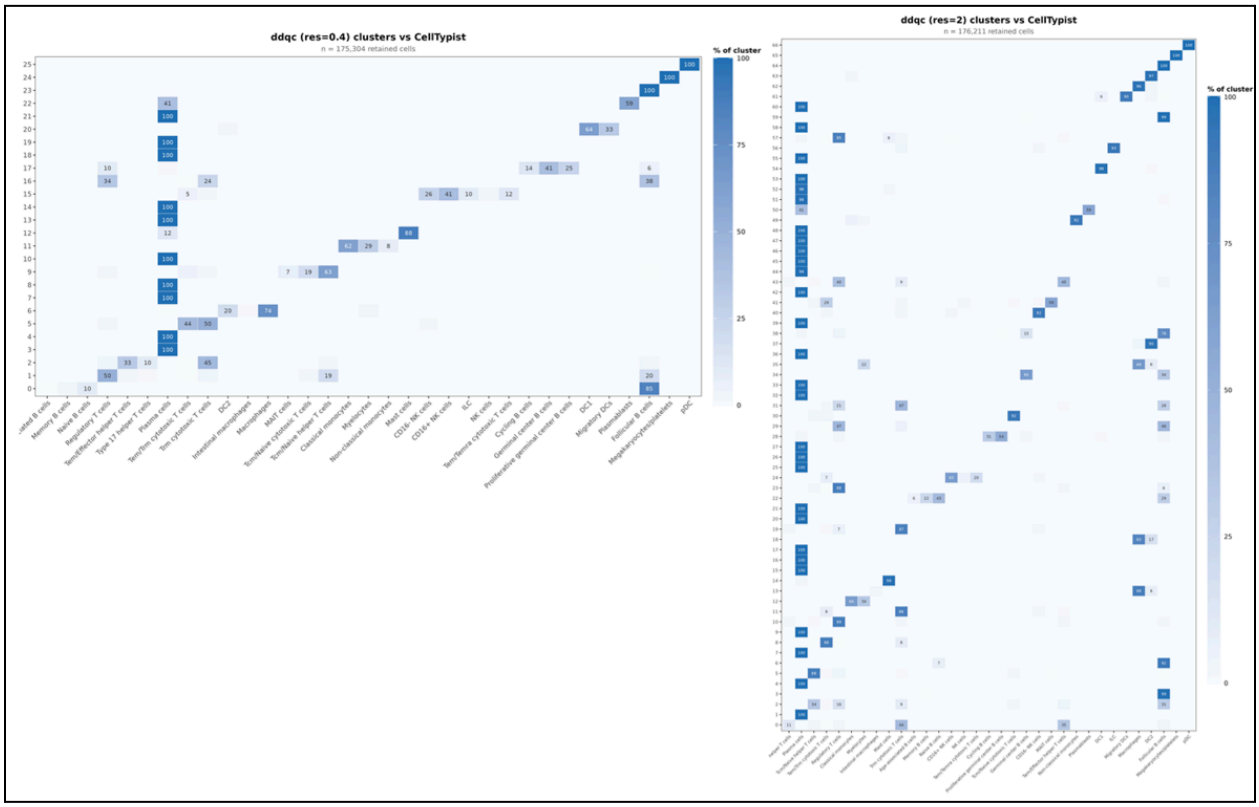

Fig 14. Heatmap generated from cross-tabulation of matching cell barcodes between ddqc clustering, resolution of 1.3, and cell type annotation using CellTypist

Discussion

Overall, we completed all planned recreations and extensions outlined in the original proposal. For the recreation, we applied ddqc, miQC, standard cutoff, and no QC to colon immune cells to evaluate the generalizability of ddqc on a different dataset. For the extensions, we compared

ddqc and miQC on a paired healthy versus disease kidney dataset, performed clustering resolution analysis for ddqc, and assessed cluster coherence using reference-based annotation tools. These extensions aimed to test the paper's assumption that cells within a cluster share similar biological properties.

However, several limitations should be acknowledged. First, there is a discrepancy in the Louvain resolution used for ddqc: the paper reports a resolution of 1.4, while the GitHub implementation uses 1.3. We used 1.4 for the recreation and 1.3 for the resolution and annotation analyses, which may introduce slight differences in clustering patterns.

Second, our evaluation was limited to the colon immune dataset and a single disease dataset. While these provided consistent and biologically meaningful results, they do not capture the full diversity of scRNA-seq data, potentially limiting the generalizability of our findings.

Third, miQC was implemented using the Bioconductor version rather than the exact setup described in the paper. Although the underlying methodology is consistent, this may lead to differences in filtering behavior and limit direct comparability with reported results.

Another extension would be to reduce ddqc's sensitivity to clustering resolution by aggregating filtering decisions across multiple resolutions to mitigate reliance on a single clustering configuration. For example, running ddqc at several resolution values, and retaining and removing cells based on consensus across runs can provide a more robust estimate of cell quality.

Finally, an additional extension would be to evaluate clustering coherence using canonical marker genes for colon immune cell types as an internal biological reference. This was not implemented due to the lack of a standardized marker set for the dataset and the context-dependent nature of cell identity. Incorporating marker-based validation could strengthen conclusions by providing an orthogonal line of evidence, though it would still require careful interpretation, as marker expression varies across conditions and may not represent absolute ground truth.

Team Contributions

Maha Attique was responsible for drafting the Introduction, recreation related to no-QC, performing the ddqc clustering resolution analysis, and editing and proofreading the report. Naila Mawardi contributed to the recreation of standard QC, performed clustering annotation extension, and assisted with overall report proofreading and cleanup. Shriya Jain carried out the ddqc recreation using the colon dataset, contributed to the ddqc analysis to a disease versus non-disease kidney dataset, contributed to the Discussion section, and organized the report. Tazein Shah was responsible for the miQC recreation using the colon dataset, applied miQC to the disease versus non-disease dataset, ran the comparison between ddqc and miQC results, and wrote up the Results for the ddqc and miQC comparison. In addition, Tazein organized the meetings, assigned roles, and reformatted + edited the report.

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